

Mechanistic Interpretability for Masked Diffusion Language Models

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Abstract

Masked diffusion language models (MDLMs) are a new direction in language modeling research, offering a different approach compared to traditional autoregressive models. This naturally leads us to question about the causality in the generations of MDLMs as compared to those of autoregressive models.

Mechanistic interpretability (MI) is a process that aims to understand the internal reasoning processes of machine learning models. Although mechanistic interpretability techniques have been widely used on autoregressive models and have gained significant insights into understanding how autoregressive models reason, they have not yet been widely applied to diffusion-based language models.

This study aims to utilize MI techniques on MDLMs to investigate their inner causal mechanisms. We propose to use MI techniques such as circuit analysis, perturbation methods, and causal interventions. Through our work, we hope to gain new insights on interpretability of masked diffusion language models.

1 Literature Review

We first briefly explain diffusion models in general to better introduce the ideas of diffusion-based language models. We then explain the ideas of Mechanistic interpretability, how Mechanistic interpretability have been used on autoregressive models, and how they could be used for causal analysis of Masked Diffusion Language Models (MDLMs).

1.1 Diffusion Models

Diffusion models (Ho et al., 2020) constitute a class of generative models which learn to synthesize complex data (such as images, text, or audio) by modeling the data generation process as the reversal of a gradual noising procedure.

1.1.1 The Forward Diffusion Process

The key idea of a diffusion model is to define a Markov chain that gradually transforms structured data into pure noise over a sequence of time steps (Ho et al., 2020). Let the observed data be denoted by x_0 . The forward process generates a sequence of latent variables $\{x_t\}_{t=1}^T$ by progressively adding noise:

$$q(x_t | x_{t-1}) = \mathcal{N}(x_t; \sqrt{1 - \beta_t} x_{t-1}, \beta_t \mathbf{I})$$

Here, $\beta_t \in (0, 1)$ represents a small variance schedule that determines the magnitude of noise added at each step. As $t \rightarrow T$, the variable x_T converges to standard Gaussian noise: $x_T \sim \mathcal{N}(0, \mathbf{I})$.

The full forward process is given by:

$$q(x_{1:T} | x_0) = \prod_{t=1}^T q(x_t | x_{t-1})$$

Thanks to reparameterization, it is possible to directly sample x_t at any arbitrary time step using the following expression:

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, \quad \epsilon \sim \mathcal{N}(0, \mathbf{I})$$

where $\alpha_t = 1 - \beta_t$, and $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$.

1.1.2 The Reverse Denoising Process

The generative objective is to learn the reverse process $p_\theta(x_{t-1} | x_t)$, which denoises x_t step by step to recover a sample resembling x_0 (Ho et al., 2020):

$$p_\theta(x_{t-1} | x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \Sigma_\theta(x_t, t))$$

Parameters θ are optimized to approximate the true reverse dynamics, typically using a variational lower bound or a denoising score matching objective. In practice, a common simplification is to

predict the added noise ϵ directly and minimize the mean squared error (MSE):

$$\mathcal{L}_{\text{simple}}(\theta) = \mathbb{E}_{x_0, \epsilon, t} [\|\epsilon - \epsilon_\theta(x_t, t)\|^2]$$

1.1.3 From Diffusion to Language Models

When diffusion models are applied to discrete data types, Gaussian noise is no longer appropriate (Austin et al., 2021). Instead, the corruption process is defined via a discrete transition matrix. A widely adopted variant is *masked diffusion*, where tokens in the original sequence x_0 are gradually masked independently with a [MASK] token until the entire sequence is fully masked at $t = 1$. For $t \in (0, 1)$, the sequence x_t is partially masked, with the number of tokens masked according to a noise schedule.

The reverse diffusion process is parametrized with a mask predictor, $p(\cdot|x_t)$ that takes an input and predicts all masked tokens (M) simultaneously. It is trained using a cross-entropy loss computed only on the masked tokens:

$$\mathcal{L}(\theta) \triangleq -\mathbb{E}_{t, x_0, x_t} \left[\frac{1}{t} \sum_{i=1}^L \mathbf{1}[x_t^i = M] \log p_\theta(x_0^i | x_t) \right], \quad (1)$$

Notably, the loss in Eq. (1) has been proven to be an upper bound on the negative log-likelihood of the model distribution (Shi et al., 2024; Ou et al., 2024):

$$-\mathbb{E}_{p_{\text{data}}(x_0)} [\log p_\theta(x_0)] \leq \mathcal{L}(\theta), \quad (2)$$

making it a principled objective for generative modeling.

These models employ a masking ratio that varies randomly between 0 and 1 as opposed to masked language models (Devlin et al., 2019) that use a fixed ratio.

1.2 Mechanistic Interpretability

Mechanistic Interpretability (MI) is the practice of reverse-engineering neural networks to understand the internal reasoning mechanisms they use to compute their outputs (Rai et al., 2025).

1.2.1 Core ideas

There are three key ideas of understanding Mechanistic Interpretability (Rai et al., 2025):

- **Feature Superposition:** Neural networks often store multiple abstract features in overlapping directions within activation space. Techniques such as sparse autoencoders or dictionary learning have been proposed to disentangle these “superimposed” features for interpretability.
- **Circuit Analysis:** A *circuit* refers to a minimal subgraph of neurons or attention heads whose interaction performs a coherent sub-task (e.g., copying tokens, adding numbers). Circuits can often be isolated through interventions and activation patching.
- **Causal Mediation:** Inspired by methods from causal inference, this technique quantifies how much a specific component (e.g., layer or neuron group) mediates the effect of an input feature on the model’s output.

1.2.2 Successes in Autoregressive Models

Mechanistic interpretability has achieved significant breakthroughs when applied to autoregressive models (ARMs) like GPT-2 and GPT-3 (Rai et al., 2025):

- **Induction Heads:** Attention heads detect repeated token patterns in context and “copy” the following token accordingly—a key mechanism in in-context learning.
- **Arithmetic Circuits:** Studies have revealed that transformer models can perform symbolic arithmetic using reusable submodules, such as modular addition heads and positional rule circuits.
- **Sycophancy Circuits:** Models trained with human feedback can learn to agree with user biases. By isolating the circuits responsible, researchers identified attention pathways that encode sycophantic behavior, revealing hidden alignment failure modes.

2 Methodology

We now have a basic idea of Masked diffusion language models and Mechanistic interpretability, and we also acknowledge that MI techniques have not yet been systematically applied to MDLMs. Here, we identify the MI techniques we include and utilize in our approach, then, we propose our designed experiments.

2.1 Methodologies

The approach we propose is: perturbation-based analysis with causal intervention strategies (Rai et al., 2025). These methods aim to adapt and extend mechanistic interpretability tools that have proven effective for autoregressive models, while accounting for the unique properties of diffusion-based modeling.

2.1.1 Perturbation Analysis Framework

- **Layer-Wise Gradient Attribution:** We adapt probing tools such as TransformerLens to the encoder-only architecture of MDLMs. By computing input gradients and layer-wise attribution maps, we examine how information flows through the network during different denoising steps.

2.1.2 Causal Intervention Strategies

- **Time Embedding Interventions:** We manipulate intermediate time-step representations x_t by freezing or selectively re-initializing masked token states to examine the model’s reliance on the temporal denoising framework. This tests whether specific linguistic features (grammar, semantics, style) are established at different denoising steps.
- **Counterfactual Sampling:** We generate counterfactual sequences by altering the noise schedule (e.g., reversing or truncating it), then measure divergence in output quality and attribution maps. This helps identify “critical periods” in the denoising process that contribute most to final predictions.

3 Models

Our experimental framework is designed to evaluate the interpretability of Masked Diffusion Language Models (MDLMs) in comparison to traditional Autoregressive Models (ARMs). We employ a suite of controlled experiments to isolate key interpretability factors, including the sensitivity to perturbations, the validation of causal hypotheses, and comparisons across model architectures.

3.0.1 Baseline Models

To ensure comprehensive evaluation, we compare a MDLM model against a baseline large autoregressive language model. The models used in our experiments include:

- **LLaDA (Large Language Diffusion Models):** We experiment with the smaller 1B-parameter variant of LLaDA (Nie et al., 2025). This model is trained on the masked diffusion objective, where the model progressively denoises corrupted sequences with a parametrized mask predictor. We hypothesize that the iterative nature of denoising will give LLaDA a distinctive architecture compared to ARMs and hence different mechanisms of processing information.
- **Autoregressive Models (ARMs):** We propose Llama-3.2.-1B, which are typical representatives of autoregressive architectures to serve as a baseline in this study. This models generates output one token at a time, conditioned on previously generated tokens, and their interpretability has been widely studied using techniques like attention analysis and gradient attribution (Rai et al., 2025).

We plan to evaluate these models across a variety of tasks, namely natural language generation, mathematical reasoning, and code generation.

4 Experiments

4.1 Bidirectional Information Flow Analysis

Diffusion language models are claimed to have superior performance in tasks which require understanding of the entire sequence context. The iterative denoising of the entire sequence together is claimed to account for conditioning on bidirectional context instead of unidirectional attention in autoregressive models. These models have been shown to perform better on reversal tasks compared to autoregressive models with causal attention (Nie et al., 2025).

We plan to systematically study the performance of both masked diffusion and autoregressive models on such tasks and quantify the influence of bidirectional attention in diffusion models to their performance.

The following tasks will be covered in this study:

4.1.1 Text Infilling and Revision

Claim: Autoregressive models struggle with bidirectional coherence since they only have access to left context during generation, while diffusion models refine the entire sequence simultaneously.

- **Task:** Inserting or modifying text in the middle of existing content while maintaining consistency with both left and right contexts
- **Dataset:** ROCStories Corpus: This dataset consists of 5-sentence stories aimed to measure narrative coherence
- **Experiments:** A subset of the input sequence will be randomly masked and the task will be to predict the masked tokens. The following variations will be considered:
 - Position variants: Beginning / middle / end of passages
 - Length variants: 10%, 20%, 30% of tokens masked
 - Pattern variants: Single span vs. multiple non-contiguous spans
 - Single span: Mask 10%, 25%, or 50% of consecutive tokens
 - Multi-span: Mask 3-5 non-adjacent spans
- **Evaluation**
 - Perplexity of generated infills
 - Semantic coherence with surrounding context (embedding similarity at infill boundaries)
 - Grammatical errors across infill boundaries
 - ROUGE/BLEU against original masked content
 - Human evaluation of coherence and naturalness
- **Attention Analysis:** The attention maps across different layers and the masked tokens will be qualitatively and quantitatively analyzed at each diffusion step to measure context utilization and information flow in both directions.

4.2 Structured Output Generation

Claim: Diffusion models can enforce structural validity across the entire output, while autoregressive models risk accumulating structural errors that lead to invalid outputs.

- **Task:** Generating text with specific structural requirements (forms, tables, formatted text).

- **Dataset:** GitHub Code Corpus contains samples natural language text to code snippets which can be used to enforce syntax constraints.
- **Experiments:** For every code sample, an identified syntax constraint will be unmasked and the language model will be prompted to generate a program satisfying the constraints.
- **Evaluation:** Structural validity (parsability rate), Error localization and severity, Structure complexity measures (AST depth, branching factor), Human evaluation of structure and content quality
- **Analysis:** We plan to analyze the attention patterns for tokens which enforce syntax constraints. Additionally, we would be interested in the emergence of said syntactical constraints across the denoising steps in the diffusion language model.

4.3 Denoising Step Attribution

Masked Diffusion language models introduce an additional axis of information flow, the temporal denoising of the sequence. To isolate the effects of temporal denoising, we aim to conduct experiments to manipulate the intermediate partially noised representations to attribute parts of the generated samples to specific denoising time-steps.

- **Dataset:** We plan to evaluate LLaDA across natural language generation (ROCStories Corpus), mathematical reasoning (GSM-8K), and syntactical code generation (GitHub Code Corpus).
- **Experiments:**
 - Step skipping: Omit individual steps and measure quality degradation
 - Step substitution: Replace specific steps with random sampling
 - Step freezing: After specific steps, lock certain tokens and only allow others to update
- **Evaluation:** Perplexity, Semantic similarity, and human evaluations of outputs
- **Analysis:**
 - Create attribution curves showing each step’s contribution to quality

- Identify critical transition points in the diffusion process
- Determine which linguistic features (grammar, semantics, style) are established at different stages

5 Expected Outcome

We expect this study to produce novel insights into the internal mechanisms of masked diffusion language models (MDLMs) and how they differ from autoregressive models (ARMs). Specifically, our anticipated outcomes include:

- **Information Flow:** Tracing information flow with bidirectional attention in MDLMs as opposed to sequential information flow in autoregressive models.
- **Temporal Decomposition of Learning:** Evidence of temporal specialization in the denoising process, showing how different types of information (e.g., local vs. global coherence) emerge at different steps.
- **Comparison with ARMs:** A systematic contrast between the interpretability of MDLMs and ARMs, including which mechanisms are shared, altered, or unique to the masked diffusion framework.
- **Tooling Extensions:** Adaptation of tools like TransformerLens to encoder-style diffusion models, with code releases and documentation for reproducibility.

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